

Appendix A: Register Field Types

Changes from RP2040

Register field types are unchanged.

Standard types

RW:

- Read/Write
- Read operation returns the register value
- Write operation updates the register value

RO:

- Read-only
- Read operation returns the register value
- Write operations are ignored

WO:

- Write-only
- Read operation returns 0
- Write operation updates the register value

Clear types

SC:

- Self-Clearing
- Writing a 1 to a bit in an SC field will trigger an event, once the event is triggered the bit clears automatically
- Writing a 0 to a bit in an SC field does nothing

WC:

- Write-Clear
- Writing a 1 to a bit in a WC field will write that bit to 0

- Writing a 0 to a bit in a WC field does nothing
- Read operation returns the register value

FIFO types

These fields are used for reading and writing data to and from FIFOs. Accompanying registers provide FIFO control and status. There is no fixed format for the control and status registers, as they are specific to each FIFO interface.

RWF:

- Read/Write FIFO
- Reading this field returns data from a FIFO
 - When the read is complete, the data value is removed from the FIFO
 - If the FIFO is empty, a default value will be returned; the default value is specific to each FIFO interface
- Data written to this field is pushed to a FIFO, Behaviour when the FIFO is full is specific to each FIFO interface
- Read and write operations may access different FIFOs

RF:

- Read FIFO
- Functions the same as RWF, but read-only

WF:

- Write FIFO
- Functions the same as RWF, but write-only